

Pi SDR Academy



How to use the lab — student playbook & instructor notes

Offline dojo for systems, DSP, and software-defined radio on a Raspberry Pi (or Mac for development). You earn **belts** by finishing **dojos** in order. Day-to-day scoring is `./check` — you do not need to type `academy ...` for normal play.

1. One-time setup (Mac)

```
cd /Users/noamyakar/Desktop/pi-sdr-academy
source .venv/bin/activate
export ACADEMY_CURRICULUM=$PWD/curriculum
export PYTHONPATH=$PWD/platform

# Optional – installs short commands on PATH: next, hint, submit, status, dojos
pip install -e ./platform
```

On a prepared Pi image, instead source `/opt/academy/academy.env` (or your install path) after `offline_bundle/install.sh`.

2. Everyday student loop

```
python -m academy          # home screen + belt status
# or: dojos

python -m academy next      # load next challenge
cd ~/challenge              # or /challenge on Pi
# ... solve the challenge ...

./check                     # grade + score in one step
python -m academy next      # continue
```

If you found a flag in the files

```
./submit 'flag{...}'        # quotes required in zsh
```

Useful local commands (inside the challenge folder)

Command	What it does
<code>./check</code>	Runs the grader and submits on success
<code>./submit 'flag{...}'</code>	Submit a flag you found yourself
<code>./hint</code>	Reveal the next hint
<code>./next</code>	Load the next challenge

Useful global commands (after `pip install -e ./platform`)

Command	What it does
<code>next</code>	Start next unlocked challenge
<code>hint</code>	Get a hint
<code>submit</code>	Score (auto-runs check if needed)
<code>dojos / status</code>	Belt map / scoreboard
<code>python -m academy where</code>	Print challenge folder path

3. Dojo map in the browser

```
python -m academy serve --port 8081
# open http://127.0.0.1:8081/
```

- Click a dojo card to see modules, every challenge, and a short description.
- Core dojos unlock in belt order (white → black).
- Side quests are optional and do not block the core path.

4. Belts (core path)

Belt	Dojo	Focus
White	Intro to the Lab	Terminal, files, first flags
Yellow	Software Craft	Python, C/C++, GDB, CMake, Git
Orange	Systems Core	Processes, IPC, architecture
Green	DSP Foundations	Sampling, FFT, filters
Blue	Radio Path	IQ, GNU Radio, BPSK/QPSK, sync
Purple	Receivers & Radio Security	Full RX, protocol RE, CTF
Black	Capstone — Blackout	End-to-end recovery

Side quest: **Language Qualification** (brown) — Python / C++ / GNU Radio C++ depth.

5. How flags work

- Each `next` / start mints a **session flag** with a random suffix (e.g. `flag{name_a1b2c3}`).
- The flag file stays empty until you earn it via `./check` (or you paste a found flag).
- Checkers do not hardcode the live flag. Do not distribute instructor solutions to students.

6. Instructor guides (password protected)

Keep unlocked instructor material off student machines and Pi images.

```
# Unlock (password from env, prompt, or instructor/.instructor_password)
python3 scripts/unlock_instructor_guides.py
open instructor/INDEX.md

# Lock again before sharing the repo / imaging a Pi
python3 scripts/lock_instructor_guides.py
```

After editing challenges, regenerate then re-lock:

```
python3 scripts/generate_instructor_guides.py
python3 scripts/lock_instructor_guides.py
```

7. Project map

Path	Purpose
curriculum/	Modules, challenges, <code>dojos.yaml</code>
platform/	Engine, CLI, dojo web UI
<code>~/challenge</code> or <code>/challenge</code>	Active challenge workspace
<code>instructor/guides.encrypted</code>	Encrypted solution pack
<code>offline_bundle/</code>	Pi offline installer
<code>docs/</code>	Topic notes + this guide

8. Raspberry Pi (later)

1. Copy the repo (and offline bundle) to the Pi.
2. Run `offline_bundle/install.sh`.
3. `source` the generated `academy.env`.
4. Same student loop: `next` → work in `/challenge` → `./check`.
5. Do **not** unlock instructor guides on student images.

9. Quick troubleshooting

Problem	Fix
<code>next</code> / <code>academy</code> not found	Activate venv; set <code>PYTHONPATH</code> or <code>pip install -e ./platform</code>
Wrong folder	Run <code>python -m academy where</code> and <code>cd</code> there
<code>./submit flag{...}</code> fails in zsh	Use quotes: <code>./submit 'flag{...}'</code>
Old flag rejected	Session nonce changed — use planted flag or re-run <code>./check</code>
Port 8080 in use	<code>python -m academy serve --port 8081</code>
Cannot unlock guides	Set <code>INSTRUCTOR_PASSWORD</code> or check <code>instructor/.instructor_password</code>